

Extremality shift of rotating black holes at finite Gauss–Bonnet coupling

A research project carried out with AI agents

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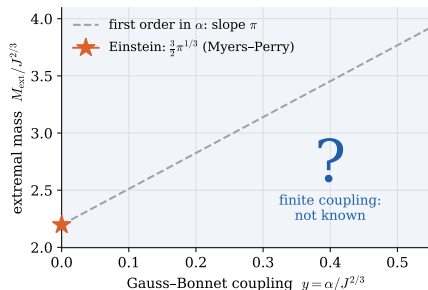
With AI assistance: **Claude Code** (Anthropic; Claude Opus 5, Sonnet 5) and **Codex** (OpenAI; GPT-5)

Gravitational Waves and AI-Assisted Research — M. Zaldarriaga & N. Mirón-Granese

Departamento de Física, FCEN, Universidad de Buenos Aires · 1–2 October 2026 ·

github.com/ddblanco/BHs

The question: how far does Gauss–Bonnet move the extremal bound?



Warm-up: Kerr, 4D ($G = c = 1$, $a = J/M$)

Extremality: $J \leq M^2$. At $J = M^2$ the horizons merge and $T = \sqrt{M^2 - a^2}/(4\pi M r_+) = 0$.

First law: $dM = T dS + \Omega_H dJ$, with $T = \kappa/2\pi$, $S = A/4$.

Einstein, 5D, two equal spins J :

$$M \geq \frac{3}{2} \pi^{1/3} J^{2/3}$$

saturated by an **extremal** ($T = 0$) black hole.

Add Gauss–Bonnet:

$$I = \frac{1}{16\pi G} \int \sqrt{-g} (R + \alpha \mathcal{L}_{\text{GB}})$$

$$\mathcal{L}_{\text{GB}} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$$

schematically $R \sim g \partial^2 g + \partial g \partial g$, $\mathcal{L}_{\text{GB}} \sim R^2$

Known: slope π at first order in α

Ma, Li, Lü, arXiv:2009.00015

Asked: the slope $\partial M_{\text{ext}}/\partial \alpha|_J$

at **finite** α .

How it started: the first prompts

“El proyecto consiste en utilizar agentes de IA para implementar códigos numéricos que permitan obtener soluciones de agujeros negros rotantes en dimensión $D > 4$ [...] reproducir soluciones numéricas ya encontradas en la literatura. [...] el paso siguiente sería emprender la búsqueda de nuevas soluciones.”

Prompt 1 — 4 Sept, 19:16, to Codex · then “Fijate que el proyecto realmente siga las indicaciones propuestas por el docente del curso” (3), “ejecutar con subagentes” (5), “continuemos con el hito 3”, “dale, avanza” · 79 human prompts, 4–28 Sept · Claude Code builds ↔ Codex referees

```
ruta-de-trabajo-inicial.txt 5 Sept 2026
52 ## Hito 4 – EGB rotante asintóticamente plano
53
54 *Salida mínima exitosa del proyecto:* familia parcial, pero validada, de
55 soluciones D=5 con momentos angulares iguales que reproduzca resultados de
56 Brihaye, Kleihaus, Kunz y Radu (2010).
57
58 - Continuar primero en velocidad angular y luego en alpha.
59 - Comparar magnitudes adimensionales y curvas publicadas.
60 - Comprobar regularidad, cargas, temperatura, entropía y primera ley.
61 - Documentar zonas donde el método pierde convergencia sin interpretarlas de
62 inmediato como fronteras físicas.
63 ... (Hito 5 – método híbrido asistido por IA)
64 ## Hito 6 – Exploración nueva
65
66
67
68 *Salida ambiciosa:* evidencia numérica reproducible sobre una extensión no
69 cubierta por los benchmarks.
70
71
72
73 Sólo se seleccionará después del Hito 4. Candidatos: otra densidad de Lovelock,
74 otra condición asintótica o una región no explorada del espacio de parámetros.
75 El spin desigual o un ansatz de cohomogeneidad 2 se consideran proyectos de
76 segunda generación por requerir PDEs.
```

first roadmap, 5 Sept. Hito 6 became Ψ ; Hito 8 ($T \rightarrow 0$, paper, referees) was not in the plan.

4.1.3 Thermodynamical properties

Considering the thermodynamics of these solutions, the EGB black holes should satisfy the first law of thermodynamics

$$dE = T_H dS + 2\Omega_H dJ. \quad (4.3)$$

One may regard the parameters S , J as a complete set of extensive parameters for the mass-energy $E(S, J)$ and define the intensive parameters conjugate to them. These quantities are the temperature and the angular velocities. Also, for $\alpha \neq 0$, these solutions do not appear to satisfy any simple Smarr relation¹⁴.

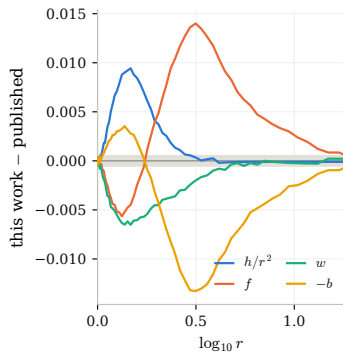
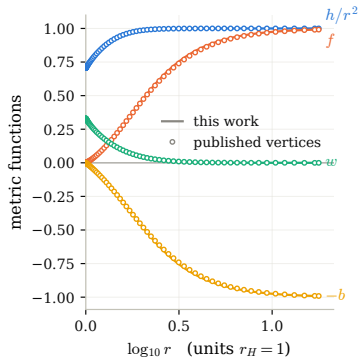
¹⁴The Smarr relation in Einstein gravity is $E = 3(T_H S + 2\Omega_H J)/2$. The recent work [43] proposes a Smarr relation for static black holes in Lovelock gravity. It would be interesting to extend the formalism in [43] to the case of spinning solutions.

arXiv:1010.0860, §4.1.3 and footnote 14.

Hito 6, α as a thermodynamic variable:

$$2M = 3TS + 6\Omega_H J + 2\alpha\Psi$$

Step 1: reproduce the published family (arXiv:1010.0860, fig. 1)



$$2 \times 10^{-4}$$

agreement at the horizon
(a third of their plotting resolution)

$$\sim 10^{-2}$$

interior gap — but their curve is
as far from the *exact* solution

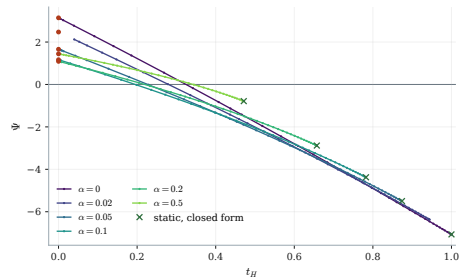
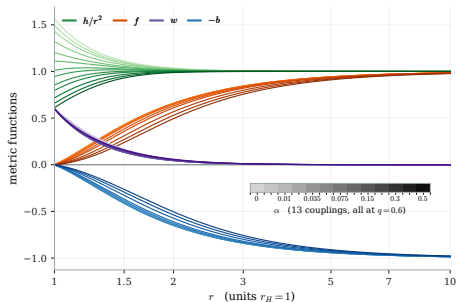
$$1.1021 \text{ vs } 1.104$$

ergosurface radius:
open, reported as such

Curves read as vectors from the arXiv PostScript source, not digitised from the image.

Step 2: measure the slope on each solution, not from differences

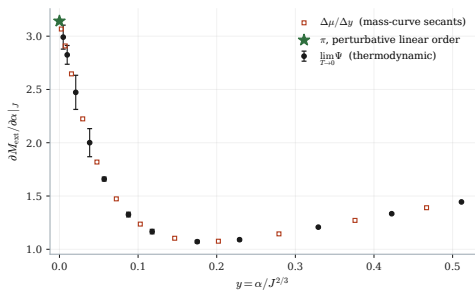
$$dM = T dS + 2\Omega_H dJ + \Psi d\alpha \quad \Rightarrow \quad \left. \frac{\partial M_{\text{ext}}}{\partial \alpha} \right|_J = \lim_{T \rightarrow 0} \Psi \quad A \frac{du}{d\alpha} = -\frac{\partial R}{\partial \alpha} \quad (\text{one linear solve})$$



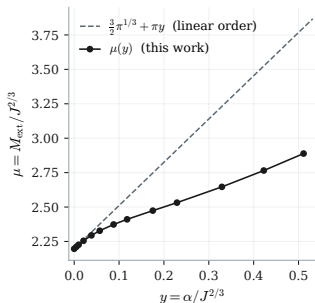
356 nonlinear solutions, **13** couplings $0 \leq \alpha \leq \frac{1}{2}$

Ψ against temperature; the slope is the value at $T = 0$

Result: the bound tightens, but not linearly



slope $\partial M_{\text{ext}}/\partial\alpha|_J$



extremal mass

> 0

everywhere

$\pi \rightarrow 1.07$

then up to 1.44

$+31\%$

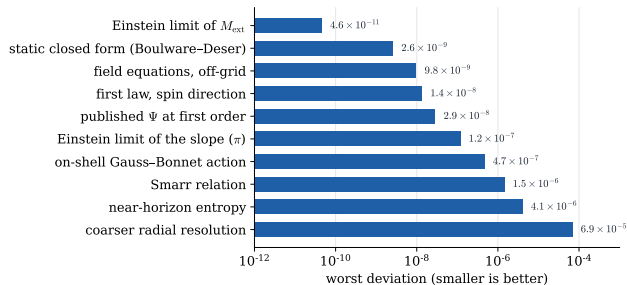
M_{ext} growth

$\times 1.32$

first-order overshoot

$T = 0$ values are extrapolations from the coldest states reached ($TJ^{1/3} \approx 3 \times 10^{-6}$).

How far can it be trusted?



Independent checks

none uses the linear-response solve.

Agents refereeing agents

a requested check *failed*; an exact bound caught a real mass error.

Still open

extrapolation to $T = 0$;
equal spins; one branch only.

What it cost

36.8 h

active agent time, 13 days

1.42 billion

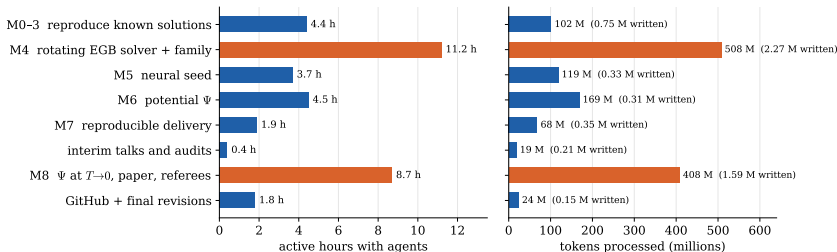
tokens processed
(99% re-read context)

5.9 million

tokens written (≈ 24 MB)

79

human prompts



The first **new** object cost $5\times$ the tokens of reproducing everything known.

Not counted: human time, background numerical runs.